MAT 022A Midterm 1

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Name: Key

SID:

Instructions: Do not turn the page until instructed to do so. No external help or calculators allowed. Please show your work (with the exception of row operations) and explain your answers if necessary. Box your answer if applicable.

1. Given $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\det A \neq 0$, find $\operatorname{adj} A$ and show that

$$A^{-1} = \begin{bmatrix} \dfrac{d}{ad-bc} & \dfrac{-b}{ad-bc} \\ \dfrac{-c}{ad-bc} & \dfrac{a}{ad-bc} \end{bmatrix}.$$

$$\operatorname{adj} A = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}^T = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\det A = ad - bc$$

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A$$

$$= \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{d}{ad-bc} & \dfrac{-b}{ad-bc} \\ \dfrac{-c}{ad-bc} & \dfrac{a}{ad-bc} \end{bmatrix}$$

2. Given the LU-factorization of A below, find $\det A$.

$$A = \begin{bmatrix} 2 & 3 & 0 & 1 \\ 4 & 5 & 3 & 3 \\ -2 & -6 & 7 & 7 \\ 8 & 9 & 5 & 21 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 3 & 1 & 0 \\ 4 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 0 & 1 \\ 0 & -1 & 3 & 1 \\ 0 & 0 & -2 & 5 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

$\det A = \det L \det U$

$\det L = 1 \begin{vmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 3 & 2 & 1 \end{vmatrix} = 1 \cdot 1 \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1 \cdot 1 \cdot 1 \cdot 1 = 1$

$\det U = 2 \begin{vmatrix} -1 & 3 & 1 \\ 0 & -2 & 5 \\ 0 & 0 & 4 \end{vmatrix} = 2(-1) \begin{vmatrix} -2 & 5 \\ 0 & 4 \end{vmatrix} = 2(-1)(-2)(4)$

$= 16$

$\det A = 1 \cdot 16 = 16$

3. Let A and B be $n \times n$ matrices. Show that if AB is invertible, then A and B are also invertible.

AB invertible means $0 \neq \det AB = \det A \det B$.

Thus $\det A \neq 0$ and $\det B \neq 0$, which means A and B are invertible.

4. Which of the following sets of vectors span $\mathbb{R}^3$?

a. $\left\{ \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} \right\}$

Does not span $\mathbb{R}^3$.

b. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Does span $\mathbb{R}^3$

c. $\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 6 \\ 4 \\ 8 \end{bmatrix} \right\}$

$$\begin{bmatrix} 1 & 2 & 5 & 6 \\ 1 & 1 & 3 & 4 \\ 2 & 2 & 6 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 5 & 6 \\ 0 & -1 & -2 & -2 \\ 0 & -2 & -4 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Does not span $\mathbb{R}^3$

5. Let $L : P_4 \to \mathbb{R}^4$ be a linear transformation defined by

$$L\left(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4\right) = \begin{bmatrix} 1 & 0 & -1 & 3 & -1 \\ 1 & 0 & 0 & 2 & -1 \\ 2 & 0 & -1 & 5 & -1 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & 0 & -1 & 3 & -1 \\ 1 & 0 & 0 & 2 & -1 \\ 2 & 0 & -1 & 5 & -1 \\ 0 & 0 & -1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

a. Find a basis for $\operatorname{ran} L$ and $\ker L$.

basis for ran L : $\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ -1 \\ 0 \end{bmatrix} \right\}$

$a_1 = s$ $\quad a_0 = -2t$

$a_3 = t$ $\quad a_2 = t$

$a_4 = 0$

$\ker L = \{ sx - 2t + tx^2 + tx^3 \mid s,t \in \mathbb{R} \}$

$= \{ sx + t(-2 + x^2 + x^3) \mid s,t \in \mathbb{R} \}$

basis for ker L : $\{ x, -2 + x^2 + x^3 \}$

b. Find $\operatorname{rank} L$ and $\operatorname{null} L$. Is $\operatorname{rank} L + \operatorname{null} L = \dim P_4$?

rank $L = 3$ $\quad$ dim $P_4 = 5$

null $L = 2$

$3 + 2 = 5$ ✓

6. Suppose that $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is a linearly independent set of vectors in a vector space V. Is $W = \{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$ where $\vec{w}_1 = \vec{v}_1 + \vec{v}_2$, $\vec{w}_2 = \vec{v}_1 + \vec{v}_3$, and $\vec{w}_3 = \vec{v}_2 + \vec{v}_3$, linearly dependent or linearly independent? Justify your answer.

$$\begin{aligned} 0 &= c_1\vec{w}_1 + c_2\vec{w}_2 + c_3\vec{w}_3 \\ &= c_1(\vec{v}_1 + \vec{v}_2) + c_2(\vec{v}_1 + \vec{v}_3) + c_3(\vec{v}_2 + \vec{v}_3) \\ &= (c_1 + c_2)\vec{v}_1 + (c_1 + c_3)\vec{v}_2 + (c_2 + c_3)\vec{v}_3 \end{aligned}$$

Since $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ are linearly independent, then

$$\begin{aligned} c_1 + c_2 \quad &= 0 \\ c_1 \quad + c_3 &= 0 \\ c_2 + c_3 &= 0 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right] \sim$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \Rightarrow \begin{array}{l} c_1 = 0 \\ c_2 = 0 \\ c_3 = 0 \end{array}$$

Thus, T is linearly independent.

7. Let $f : \mathbb{R}^n \to \mathbb{R}^m$ and $g : \mathbb{R}^m \to \mathbb{R}^n$ be linear transformations defined by $f(\vec{x}) = M\vec{x}$ and $g(\vec{y}) = M^T\vec{y}$ where M is an $m \times n$ matrix. Let $\vec{u} \in \ker f$ and $\vec{v} \in \operatorname{ran} g$. Show that $\vec{u} \cdot \vec{v} = 0$.

Hint: Let $M = \begin{bmatrix} \vec{a}_1^T \\ \vdots \\ \vec{a}_m^T \end{bmatrix}$.

$M^T = [\vec{a}_1 \cdots \vec{a}_m]$

Since $\vec{v} \in \operatorname{ran} g$, then $\vec{v}$ is in the span of the columns of M^T, i.e.

$$\vec{v} = c_1\vec{a}_1 + \cdots + c_n\vec{a}_n \text{ for } c_i \in \mathbb{R}.$$

Since $\vec{u} \in \ker f$, then $M\vec{u} = \vec{0}$ or $\begin{bmatrix} \vec{a}_1^T \\ \vdots \\ \vec{a}_m^T \end{bmatrix} \vec{u} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$, i.e.

$$\vec{a}_1^T\vec{u} = \vec{a}_1 \cdot \vec{u} = 0, \ldots, \vec{a}_m^T\vec{u} = \vec{a}_m \cdot \vec{u} = 0.$$

Thus, $\vec{u} \cdot \vec{v} = \vec{u} \cdot (c_1\vec{a}_1 + \cdots + c_n\vec{a}_n)$

$$= c_1 \underbrace{\vec{u} \cdot \vec{a}_1}_{=0} + \cdots + c_n \underbrace{\vec{u} \cdot \vec{a}_n}_{=0}$$

$$= 0.$$